

6.4.1 Numerical stability of triangular solve

6.4.1 Backward error of triangular solve, Part 1

fit width



We now use the error results for the dot product to derive a backward error result for solving $Lx = y$, where L is an $n \times n$ lower triangular matrix, via the algorithm in [Figure 6.4.1.1](#), a variation on the algorithm in [Figure 5.3.5.1](#) that stores the result in vector x and does not assume that L is unit lower triangular.

Solve $Lx = y$

$$L \rightarrow \left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right), x \rightarrow \left(\begin{array}{c} x_T \\ x_B \end{array} \right), y \rightarrow \left(\begin{array}{c} y_T \\ y_B \end{array} \right)$$

L_{TL} is 0×0 and y_T, x_T have 0 elements

while $n(L_{TL}) < n(L)$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \rightarrow \left(\begin{array}{cc|c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} x_T \\ x_B \end{array} \right) \rightarrow \left(\begin{array}{c} x_0 \\ \chi_1 \\ x_2 \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \rightarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$$

$\chi_1 := (\psi_1 - l_{10}^T x_0) / \lambda_{11}$

$$\left(\begin{array}{c|c} L_{TL} & L_{TR} \\ \hline L_{BL} & L_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} L_{00} & l_{01} & L_{02} \\ \hline l_{10}^T & \lambda_{11} & l_{12}^T \\ L_{20} & l_{21} & L_{22} \end{array} \right), \left(\begin{array}{c} x_T \\ x_B \end{array} \right) \leftarrow \left(\begin{array}{c} x_0 \\ \chi_1 \\ x_2 \end{array} \right), \left(\begin{array}{c} y_T \\ y_B \end{array} \right) \leftarrow \left(\begin{array}{c} y_0 \\ \psi_1 \\ y_2 \end{array} \right)$$

endwhile

Figure 6.4.1.1. Dot product based lower triangular solve algorithm.

To establish the backward error result for this algorithm, we need to understand the error incurred in the key computation

$$\chi_1 := (\psi_1 - l_{10}^T x_0) / \lambda_{11}.$$

The following theorem gives the required (forward error) result, abstracted away from the specifics of how it occurs in the lower triangular solve algorithm.

Lemma 6.4.1.2. Let $n \geq 1$, $\lambda, \nu \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$. Assume $\lambda \neq 0$ and consider the computation

$$\nu := (\alpha - x^T y) / \lambda,$$

Then

$$(\lambda + \delta\lambda)\tilde{\nu} = \alpha - (x + \delta x)^T y, \text{ where } |\delta x| \leq \gamma_n |x| \text{ and } |\delta\lambda| \leq \gamma_2 |\lambda|.$$

Homework 6.4.1.1. Prove [Lemma 6.4.1.2](#)

[▶ Hint](#) [▶ Solution](#)

The error result for the algorithm in [Figure 6.4.1.1](#) is given by

Theorem 6.4.1.3. Let $L \in \mathbb{R}^{n \times n}$ be a nonsingular lower triangular matrix and let $\tilde{x}$ be the computed result when executing [Figure 6.4.1.1](#) to solve $Lx = y$ under the computation model from [Subsection 6.2.3](#). Then there exists a matrix ΔL such that

$$(L + \Delta L)\tilde{x} = y \text{ where } |\Delta L| \leq \max(\gamma_2, \gamma_{n-1})|L|.$$

The reasoning behind the result is that one expects the maximal error to be incurred during the final iteration when computing $\chi_1 := (\psi_1 - l_{10}^T x_0) / \lambda_{11}$. This fits [Lemma 6.4.1.2](#), except that this assignment involves a dot product with vectors of length $n - 1$ rather than of length n .

You now prove [Theorem 6.4.1.3](#) by first proving the special cases where $n = 1$ and $n = 2$, and then the general case.

Homework 6.4.1.2. Prove [Theorem 6.4.1.3](#) for the case where $n = 1$.

[▼ Solution](#)

Case 1: $n = 1$.

The system looks like $\lambda_{11}\chi_1 = \psi_1$ so that

$$\chi_1 = \psi_1 / \lambda_{11}$$

and

$$\tilde{\chi}_1 = \psi_1 / \lambda_{11} \frac{1}{1 + \epsilon_j}$$

Rearranging gives us

$$\lambda_{11} \tilde{\chi}_1 (1 + \epsilon_j) = \psi_1$$

or

$$(\lambda_{11} + \delta \lambda_{11}) \tilde{\chi}_1 = \psi_1$$

where $\delta \lambda_{11} = \epsilon_j \lambda_{11}$ and hence

$$\begin{aligned} |\delta \lambda_{11}| &= |\epsilon_j| |\lambda_{11}| \\ &\leq \gamma_1 |\lambda_{11}| \\ &\leq \gamma_2 |\lambda_{11}| \\ &\leq \max(\gamma_2, \gamma_{n-1}) |\lambda_{11}|. \end{aligned}$$

Homework 6.4.1.3. Prove [Theorem 6.4.1.3](#) for the case where $n = 2$.

▼ [Solution](#)

Case 2: $n = 2$.

The system now looks like

$$\left(\begin{array}{c|c} \lambda_{00} & 0 \\ \hline \lambda_{10} & \lambda_{11} \end{array} \right) \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}.$$

From the proof of Case 1 we know that

$$(\lambda_{00} + \delta \lambda_{00}) \tilde{\chi}_0 = \psi_0, \text{ where } |\delta \lambda_{00}| \leq \gamma_1 |\lambda_{00}|. \quad (6.4.1)$$

Since $\chi_1 = (\psi_1 - \lambda_{10} \tilde{\chi}_0) / \lambda_{11}$, [Lemma 6.4.1.2](#) tells us that

$$(\lambda_{10} + \delta \lambda_{10}) \tilde{\chi}_0 + (\lambda_{11} + \delta \lambda_{11}) \tilde{\chi}_1 = \psi_1, \quad (6.4.2)$$

where

$$|\delta \lambda_{10}| \leq \gamma_1 |\lambda_{10}| \text{ and } |\delta \lambda_{11}| \leq \gamma_2 |\lambda_{11}|.$$

[\(6.4.1\)](#) and [\(6.4.2\)](#) can be combined into

$$\left(\begin{array}{c|c} \lambda_{00} + \delta \lambda_{00} & 0 \\ \hline \lambda_{10} + \delta \lambda_{10} & \lambda_{11} + \delta \lambda_{11} \end{array} \right) \begin{pmatrix} \tilde{\chi}_0 \\ \tilde{\chi}_1 \end{pmatrix} = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix},$$

where

$$\left(\begin{array}{c|c} |\delta \lambda_{00}| & 0 \\ \hline |\delta \lambda_{10}| & |\delta \lambda_{11}| \end{array} \right) \leq \left(\begin{array}{c|c} \gamma_1 |\lambda_{00}| & 0 \\ \hline \gamma_1 |\lambda_{10}| & \gamma_2 |\lambda_{11}| \end{array} \right).$$

Since $\gamma_1 \leq \gamma_2$

$$\left| \left(\begin{array}{c|c} \delta \lambda_{00} & 0 \\ \hline \delta \lambda_{10} & \delta \lambda_{11} \end{array} \right) \right| \leq \max(\gamma_2, \gamma_{n-1}) \left| \left(\begin{array}{c|c} \lambda_{00} & 0 \\ \hline \lambda_{10} & \lambda_{11} \end{array} \right) \right|.$$

Homework 6.4.1.4. Prove [Theorem 6.4.1.3](#) for $n \geq 1$.

▼ [Solution](#)

We will utilize a proof by induction.

- Case 1: $n = 1$.

See [Homework 6.4.1.2](#).

- Case 2: $n = 2$.

See [Homework 6.4.1.3](#).

- Case 3: $n > 2$.

The system now looks like

$$\left(\begin{array}{c|c} L_{00} & 0 \\ \hline l_{10}^T & \lambda_{11} \end{array} \right) \begin{pmatrix} x_0 \\ \chi_1 \end{pmatrix} = \begin{pmatrix} y_0 \\ \psi_1 \end{pmatrix}, \quad (6.4.3)$$

where $L_{00} \in \mathbb{R}^{(n-1) \times (n-1)}$, and the inductive hypothesis states that

$$(L_{00} + \delta L_{00}) \tilde{x}_0 = y_0 \text{ where } |\delta L_{00}| \leq \max(\gamma_2, \gamma_{n-2}) |L_{00}|.$$

Since $\chi_1 = (\psi_1 - l_{10}^T \tilde{x}_0) / \lambda_{11}$, [Lemma 6.4.1.2](#) tells us that

$$(l_{10} + \delta l_{10})^T \tilde{x}_0 + (\lambda_{11} + \delta \lambda_{11}) \tilde{\chi}_1 = \psi_1, \quad (6.4.4)$$

where $|\delta l_{10}| \leq \gamma_{n-1} |l_{10}|$ and $|\delta \lambda_{11}| \leq \gamma_2 |\lambda_{11}|$.

[\(6.4.3\)](#) and [\(6.4.4\)](#) can be combined into

$$\left(\begin{array}{c|c} L_{00} + \delta L_{00} & 0 \\ \hline (l_{10} + \delta l_{10})^T & \lambda_{11} + \delta \lambda_{11} \end{array} \right) \begin{pmatrix} \tilde{x}_0 \\ \tilde{\chi}_1 \end{pmatrix} = \begin{pmatrix} y_0 \\ \psi_1 \end{pmatrix},$$

where

$$\left(\begin{array}{c|c} |\delta L_{00}| & 0 \\ \hline |\delta l_{10}^T| & |\delta \lambda_{11}| \end{array} \right) \leq \left(\begin{array}{c|c} \max(\gamma_2, \gamma_{n-2}) |L_{00}| & 0 \\ \hline \gamma_{n-1} |l_{10}^T| & \gamma_2 |\lambda_{11}| \end{array} \right)$$

and hence

$$\left| \left(\frac{\delta L_{00}}{\delta l_{10}^T} \middle| \frac{0}{\delta \lambda_{11}} \right) \right| \leq \max(\gamma_2, \gamma_{n-1}) \left| \left(\frac{L_{00}}{l_{10}^T} \middle| \frac{0}{\lambda_{11}} \right) \right|.$$

- By the Principle of Mathematical Induction, the result holds for all $n \geq 1$.

6.4.1 Backward error ltrsv, Part 2

fit width



A careful examination of the solution to [Homework 6.4.1.2](#), together with the fact that $\gamma_{n-1} \leq \gamma_n$ allows us to state a slightly looser, but cleaner, result of [Theorem 6.4.1.3](#):

Corollary 6.4.1.4. *Let $L \in \mathbb{R}^{n \times n}$ be a nonsingular lower triangular matrix and let $\tilde{x}$ be the computed result when executing [Figure 6.4.1.1](#) to solve $Lx = y$ under the computation model from [Subsection 6.2.3](#). Then there exists a matrix ΔL such that*

$$(L + \Delta L)\tilde{x} = y \text{ where } |\Delta L| \leq \gamma_n |L|.$$